

Assignment – 4

Q1. Given a range of first 10 numbers, Iterate from start number to the end number and print the sum of the current number and previous number.

```
for i in range(10):
    if i > 0:
        print("Sum at index", i, ":", 2 * i - 1)
    else:
        print("Sum at index", i, ":", i)
```

Output:

```
Sum at index 0 : 0
Sum at index 1 : 1
Sum at index 2 : 3
Sum at index 3 : 5
Sum at index 4 : 7
Sum at index 5 : 9
Sum at index 6 : 11
Sum at index 7 : 13
Sum at index 8 : 15
Sum at index 9 : 17
```

Q2. Given a string, display only those characters which are present at an even index number.

```
string = input("Enter a string: ")
print("Characters at even index are:", string[::2])
```

Output:

```
Enter a string: abcdefghijklmnopqrstuvwxyz
Characters at even index are: acegikmoqrtvqz
```

Q3. Given a string and an integer number n, remove characters from a string starting from zero up to n and return a new string.

```
string = input("Enter a string: ")
n = int(input("Enter a number: "))
print("String after removing characters upto index", n, "is:", string[n:])
```

Output:

```
Enter a string: abcdefghijklmnopqrstuvwxyz
Enter a number: 15
String after removing characters upto index 15 is: pqrstuvwxyz
```

Q4. Given a list of numbers, return True if first and last number of a list is same.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
print(lst[0] == lst[-1])
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

Output:

```
Enter a list of numbers(comma-separated): 1, 2, 3, 4, 5, 1
True
```

Q5. Given a list of numbers, Iterate it and print only those numbers which are divisible of 5.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
print("Numbers divisible by 5 are:", [element for element in lst if element % 5 == 0])
```

Output:

```
Enter a list of numbers(comma-separated): 1, 5, 9, 10, 15, 14, 20
Numbers divisible by 5 are: [5, 10, 15, 20]
```

Q6. Return the total count of sub-string appears in the given string.

```
string = input("Enter a string: ")
sub_string = input("Enter a sub string: ")
print("Total number of times the sub string appears is:", string.count(sub_string))
```

Output:

```
Enter a string: sub_string
Enter a sub string: string
Total number of times the sub string appears is: 1
```

Q7. Given a two list of numbers create a new list such that new list should contain only odd numbers from the first list and even numbers from the second list.

```
lst1 = list(eval(input("Enter the first list of numbers(comma-separated): ")))
lst2 = list(eval(input("Enter the second list of numbers(comma-separated): ")))
res = ([element for element in lst1 if element % 2 != 0] +
       [element for element in lst2 if element % 2 == 0])
print(res)
```

Output:

```
Enter the first list of numbers(comma-separated): 1, 2, 3, 4, 5
Enter the second list of numbers(comma-separated): 6, 7, 8, 9, 10
[1, 3, 5, 6, 8, 10]
```

Q8. Given a list iterate it and display numbers which are divisible by 5 and if you find number greater than 150 stop the loop iteration.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
print("Numbers divisible by 5 are:", end=" ")
```

```
for element in lst:
    if element > 150:
        break
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

```

if element % 5 == 0:
    print(element, end=" ")

```

Output:

Enter a list of numbers(comma-separated): 1, 5, 10, 99, 160, 170, 156
 Numbers divisible by 5 are: 5 10

Q9. Use a loop to display elements from a given list which are present at even positions.

```

lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
print("Elements at even positions are:", end=" ")

```

```

for i in range(len(lst)):
    if i % 2 == 0:
        print(lst[i], end=" ")

```

Output:

Enter a list of numbers(comma-separated): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
 Elements at even positions are: 1 3 5 7 9

Q10. Find the sum of the series 2 + 22 + 222 + 2222 + ... n terms.

```

n = int(input("Enter a number: "))
res = current = 0

for i in range(n):
    current = current * 10 + 2
    res += current

print("The sum of the series is:", res)

```

Output:

Enter a number: 7
 The sum of the series is: 2469134

Q11. Given a two list. Create a third list by picking an odd-index element from the first list and even index elements from second.

```

lst1 = list(eval(input("Enter the first list of numbers(comma-separated): ")))
lst2 = list(eval(input("Enter the second list of numbers(comma-separated): ")))
print("Resultant list is:", lst1[1::2] + lst2[::2])

```

Output:

Enter the first list of numbers(comma-separated): 1, 2, 3, 4, 5
 Enter the second list of numbers(comma-separated): 6, 7, 8, 9, 10
 Resultant list is: [2, 4, 6, 8, 10]

Name: Asmit Ayush
 SIC No: 24BCSE59
 Lab Roll No: 21

Q12. Given an input list removes the element at index 4 and add it to the 2nd position and also, at the end of the list.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))

if len(lst) >= 5:
    element = lst.pop(4)
    lst.insert(2, element)
    lst.append(element)
    print("Resultant list:", lst)
else:
    print("Length of the list must be greater than 4")
```

Output:

Enter a list of numbers(comma-separated): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Resultant list: [1, 2, 5, 3, 4, 6, 7, 8, 9, 10, 5]

Q13. Given a list slice it into a 3 equal chunks and reverse each list.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
size = len(lst) // 3
chunks = [lst[: size], lst[size: size * 2], lst[size * 2:]]

for chunk in chunks:
    chunk.reverse()

print("The 3 equal chunks are: ", chunks)
```

Output:

Enter a list of numbers(comma-separated): 1, 2, 3, 4, 5, 6, 7, 8, 9
The 3 equal chunks are: [[3, 2, 1], [6, 5, 4], [9, 8, 7]]

Q14. Given a two list of equal size create a set such that it shows the element from both lists in the pair.

```
lst1 = list(eval(input("Enter the first list of numbers(comma-separated): ")))
lst2 = list(eval(input("Enter the second list of numbers(comma-separated): ")))
res = [(lst1[i], lst2[i]) for i in range(len(lst1))]
print(res)
```

Output:

Enter the first list of numbers(comma-separated): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Enter the second list of numbers(comma-separated): 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
[(1, 10), (2, 9), (3, 8), (4, 7), (5, 6), (6, 5), (7, 4), (8, 3), (9, 2), (10, 1)]

Q15. Given a string of odd length greater 7, return a string made of the middle three chars of a given String.

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

```
string = input("Enter a string: ")
mid = len(string) // 2
print("The middle 3 characters are: ", string[mid - 1: mid + 2])
```

Output:

```
Enter a string: abcdefg
The middle 3 characters are: cde
```

Q16. Given 2 strings, s1 and s2, create a new string by appending s2 in the middle of s1.

```
string1 = input("Enter the first string: ")
string2 = input("Enter the second string: ")
mid = len(string) // 2
print("Resultant string is:", string1[:mid] + string2 + string1[mid:])
```

Output:

```
Enter the first string: abcdef
Enter the second string: ghijkl
Resultant string is: abcghijkldef
```

Q17. Given 2 strings, s1, and s2 return a new string made of the first, middle and last character of each input string.

```
string1 = input("Enter the first string: ")
string2 = input("Enter the second string: ")
mid1, mid2 = len(string1) // 2, len(string2) // 2
print("Resultant string is:", string1[0] + string1[mid1] + string1[-1] + string2[0] + string2[mid2] + string2[-1])
```

Output:

```
Enter the first string: abcdef
Enter the second string: ghijkl
Resultant string is: adfgjl
```

Q18. Arrange string characters such that lowercase letters should come first.

```
string = input("Enter a string: ")
res = ""
```

```
for ch in string:
    if ch.islower():
        res = ch + res
    else:
        res += ch
```

```
print("Resultant string is:", res)
```

Output:

```
Enter a string: ABCdefGHIjkl
Resultant string is: lkjfedABCGHI
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

Q19. Count all lower case, upper case, digits, and special symbols from a given string.

```

string = input("Enter a string: ")
counts = [0] * 4

for ch in string:
    if ch.islower():
        counts[0] += 1
    elif ch.isupper():
        counts[1] += 1
    elif ch.isdigit():
        counts[2] += 1
    else:
        counts[3] += 1

print("Number of lower case letters are:", counts[0])
print("Number of upper case letters are:", counts[1])
print("Number of digits are:", counts[2])
print("Number of special symbols are:", counts[3])

```

Output:

```

Enter a string: Hello, I am him!!!
Number of lower case letters are: 9
Number of upper case letters are: 2
Number of digits are: 0
Number of special symbols are: 7

```

Q20. Given a string, return the sum and average of the digits that appear in the string, ignoring all other characters.

```

string = input("Enter a string: ")
res = cnt = 0

for ch in string:
    if ch.isdigit():
        res += int(ch)
        cnt += 1

print("Sum:", res)
print("Average:", res / cnt)

```

Output:

```

Enter a string: Dream2503
Sum: 10
Average: 2.5

```

Q21. Write a Python program to remove duplicates from a list

```

lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
res = []

```

Name: Asmit Ayush
 SIC No: 24BCSE59
 Lab Roll No: 21

```

for element in lst:
    if element not in res:
        res.append(element)

print("The list with unique elements are:", res)

```

Output:

Enter a list of numbers(comma-separated): 2, 5, 0, 3, 8, 6, 6, 0
 The list with unique elements are: [2, 5, 0, 3, 8, 6]

Q22. Write a Python program to find the second largest number in a list.

```

lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
max = res = lst[0]

for element in lst[1:]:
    if element > max:
        res = max
        max = element
    elif element > res:
        res = element

if res == max:
    print("There is no second maximum element in the list")
else:
    print("The second maximum element in the list is:", res)

```

Output:

Enter a list of numbers(comma-separated): 2, 5, 0, 3, 8, 6, 6, 0
 The second maximum element in the list is: 6

Q23. Write a Python program to sort a list of elements using the bubble sort algorithm.

```

lst = list(eval(input("Enter a list of numbers(comma-separated): ")))

for i in range(len(lst) - 1):
    for j in range(len(lst) - i - 1):
        if lst[j] > lst[j + 1]:
            lst[j], lst[j + 1] = lst[j + 1], lst[j]

print("The list after bubble sort is:", lst)

```

Output:

Enter a list of numbers(comma-separated): 2, 5, 0, 3, 8, 6, 6, 0
 The list after bubble sort is: [0, 0, 2, 3, 5, 6, 6, 8]

Name: Asmit Ayush
 SIC No: 24BCSE59
 Lab Roll No: 21

Q24. Write a Python program to find the common elements between two lists.

```
lst1 = list(eval(input("Enter the first list of numbers(comma-separated): ")))
lst2 = list(eval(input("Enter the second list of numbers(comma-separated): ")))
res = [element for element in lst1 if element in lst2]
print("The intersection of the two list are:", res)
```

Output:

```
Enter the first list of numbers(comma-separated): 2, 5, 0, 3
Enter the second list of numbers(comma-separated): 8, 6, 6, 0
The intersection of the two list are: [0]
```

Q25. Write a Python program to count the frequency of each element in a list.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))
seen = []
print("The frequency of all the element in the list is:", end=" ")
```

```
for element in lst:
    if element not in seen:
        print(element, "-> ", lst.count(element), sep="", end=" ")
        seen.append(element)
```

Output:

```
Enter a list of numbers(comma-separated): 2, 5, 0, 3, 8, 6, 6, 0
The frequency of all the element in the list is: 2->1 5->1 0->2 3->1 8->1 6->2
```

Q26. Write a Python program to count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings

```
lst = list(eval(input("Enter a list of strings(comma-separated): ")))
res = [element for element in lst if len(element) >= 2 and element[0] == element[-1]]
print("The resultant list is:", res)
```

Output:

```
Enter a list of strings(comma-separated): "aba", "xyz", "aa", "x", "bbb", "abca", "cc"
The resultant list is: ['aba', 'aa', 'bbb', 'abca', 'cc']
```

Q27. Write a Program that get two lists as input and check if they have at least one common member

```
lst1 = list(eval(input("Enter the first list of numbers(comma-separated): ")))
lst2 = list(eval(input("Enter the second list of numbers(comma-separated): ")))
```

```
for element in lst1:
    if element in lst2:
        print("They have one common element ie:", element)
        break
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21


```
else:
    print("They have no common element")
```

Output:

Enter the first list of numbers(comma-separated): 2, 5, 0, 3
 Enter the second list of numbers(comma-separated): 8, 6, 6, 0
 They have one common element ie: 0

Q28. Write a Python program to print a specified list after removing the 0th, 4th and 5th elements.

```
lst = list(eval(input("Enter a list of numbers(comma-separated): ")))

if len(lst) > 5:
    for idx in (5, 4, 0):
        lst.pop(idx)

    print("The resultant list is:", lst)
else:
    print("There must be atleast 6 element in the list")
```

Output:

Enter a list of numbers(comma-separated): 2, 5, 0, 3, 8, 6, 6, 0
 The resultant list is: [5, 0, 3, 6, 0]

Q29. Write a Python program to generate and print a list of first and last 5 elements where the values are square of numbers between 1 and 30

```
res = [element ** 2 for element in range(1, 6)] + [element ** 2 for element in range(25, 31)]
print("The resultant list is:", res)
```

Output:

The resultant list is: [1, 4, 9, 16, 25, 625, 676, 729, 784, 841, 900]

Q30. Write a Python Program to print all Possible Combinations from the three Digits

```
lst = input("Enter the 3 digits: ")
print("The the combination of the 3 digits,", lst, "are:")

for first in lst:
    for middle in lst:
        for last in lst:
            print(first, middle, last, sep="", end=" ")
        print()
```

Output:

Enter the 3 digits: 729
 The the combination of the 3 digits, 729 are:
 777 772 779

Name: Asmit Ayush
 SIC No: 24BCSE59
 Lab Roll No: 21

727 722 729
797 792 799
277 272 279
227 222 229
297 292 299
977 972 979
927 922 929
997 992 999